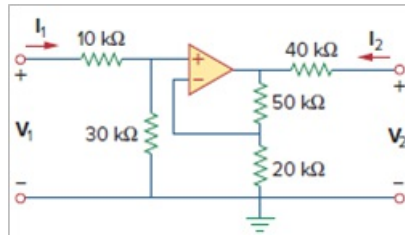


Problem

Find the z parameters of the op amp circuit in Fig. 19.109. Obtain the transmission parameters.

Figure 19.109



Step-by-step solution

Step 1 of 3

Consider the following op amp circuit with nodes marked as a and b .

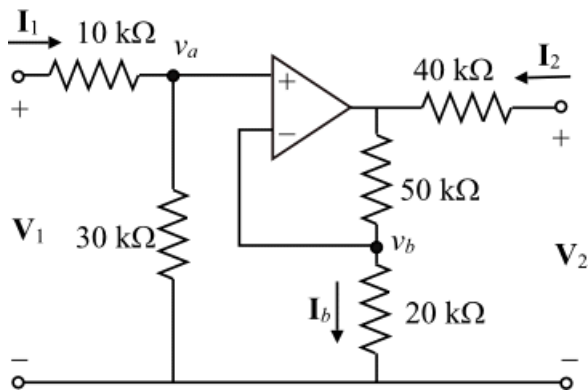


Figure 1

Step 2 of 3

Define z parameters as follows.

$$V_1 = z_{11}I_1 + z_{12}I_2$$

$$V_2 = z_{21}I_1 + z_{22}I_2$$

$$z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$$

$$z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

$$z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

To determine z_{11} and z_{21} , apply voltage source V_1 and leave the output port open.

Apply Kirchhoff's voltage law to the input loop.

$$V_1 = (10 \text{ k} + 30 \text{ k})I_1$$

$$V_1 = (40 \text{ k})I_1 \dots\dots (1)$$

Apply Kirchhoff's current law at node a .

$$\begin{aligned}\frac{V_1 - v_a}{10 \text{ k}} &= \frac{v_a}{30 \text{ k}} \\ \frac{V_1}{10 \text{ k}} &= \left(\frac{1}{30 \text{ k}} + \frac{1}{10 \text{ k}} \right) v_a \\ \frac{v_1}{10 \text{ k}} &= \frac{4}{30 \text{ k}} v_a \\ 3V_1 &= 4v_a \\ v_a &= \frac{3}{4} V_1\end{aligned}$$

But, $v_a = v_b$

Thus,

$$\begin{aligned}v_a = v_b &= \frac{3}{4} V_1 \\ &= \frac{3}{4} (40 \text{ k}) I_1 \\ &= (30 \text{ k}) I_1\end{aligned}$$

To determine z_{12} and z_{22} , apply voltage source V_2 to the output port and leave the input port open.

Apply Kirchhoff's voltage law to the output loop.

$$V_2 = (40 \text{ k}) I_2 + (50 \text{ k} + 20 \text{ k}) I_b \dots\dots (2)$$

But,

$$I_b = \frac{V_b}{20 \text{ k}}$$

Substitute $(30 \text{ k}) I_1$ for V_b .

$$\begin{aligned}I_b &= \frac{(30 \text{ k}) I_1}{20 \text{ k}} \\ &= \frac{3}{2} I_1\end{aligned}$$

Substitute $\frac{3}{2} I_1$ for I_b in equation (2).

$$\begin{aligned}V_2 &= (40 \text{ k}) I_2 + (50 \text{ k} + 20 \text{ k}) \frac{3}{2} I_1 \\ &= (40 \text{ k}) I_2 + (70 \text{ k}) \frac{3}{2} I_1 \\ V_2 &= (105 \text{ k}) I_1 + (40 \text{ k}) I_2 \dots\dots (3)\end{aligned}$$

Write z parameters from equations (1) and (3).

$$[z] = \begin{bmatrix} 40 & 0 \\ 105 & 40 \end{bmatrix} \text{ k}\Omega$$

Thus, the z parameters of the op amp circuit are $\boxed{\begin{bmatrix} 40 & 0 \\ 105 & 40 \end{bmatrix} \text{ k}\Omega}$.

Determine Δ_z using the following formula.

$$\begin{aligned}\Delta_z &= z_{11}z_{22} - z_{12}z_{21} \\ &= (40)(40) - 0 \\ &= 1600\end{aligned}$$

Convert the z parameters into T parameters using the following conversion formulae.

$$\begin{aligned}A &= \frac{z_{11}}{z_{21}} \\ B &= \frac{\Delta_z}{z_{21}} \\ C &= \frac{1}{z_{21}}\end{aligned}$$

$$\mathbf{D} = \frac{z_{22}}{z_{21}}$$

Determine **A** .

$$\begin{aligned}\mathbf{A} &= \frac{z_{11}}{z_{21}} \\ &= \frac{40}{105} \\ &= 0.381\end{aligned}$$

Determine **B** .

$$\begin{aligned}\mathbf{B} &= \frac{\Delta_z}{z_{21}} \\ &= \frac{1600}{105} \\ &= 15.238 \text{ k}\Omega\end{aligned}$$

Determine **C** .

$$\begin{aligned}\mathbf{C} &= \frac{1}{z_{21}} \\ &= \frac{1}{105 \times 10^3} \\ &= 9.524 \times 10^{-6} \text{ S}\end{aligned}$$

Determine **D** .

$$\begin{aligned}\mathbf{D} &= \frac{z_{22}}{z_{21}} \\ &= \frac{40}{105} \\ &= 0.381\end{aligned}$$

Thus, the T parameters of the op amp circuit are $\boxed{\begin{bmatrix} 0.381 & 15.238 \text{ k}\Omega \\ 9.524 \times 10^{-6} \text{ S} & 0.381 \end{bmatrix}}$.